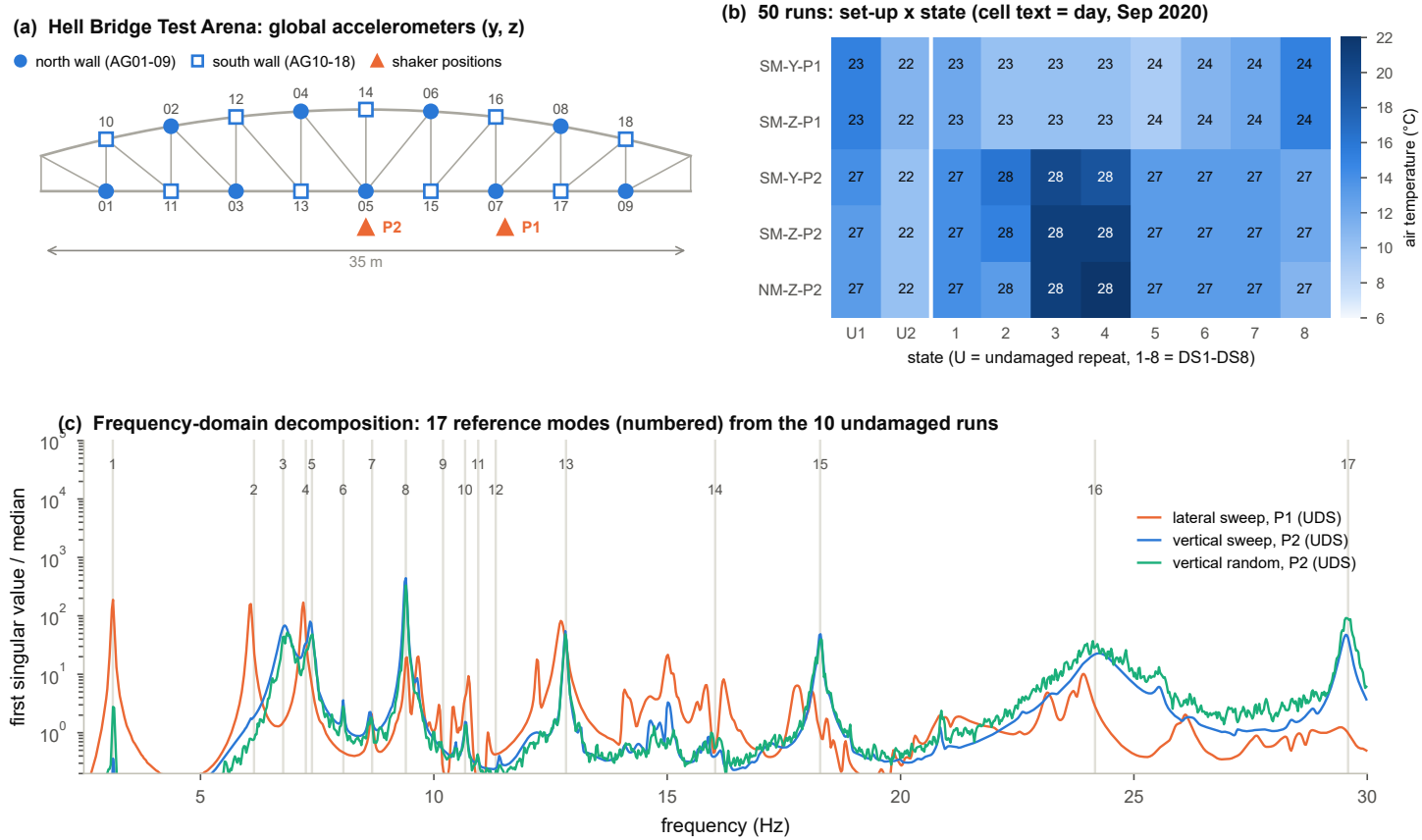

(a) Hell Bridge Test Arena: global accelerometers (y, z)
north wall (AG01-09)
south wall (AG10-18)
shaker positions
10
02
12
04
14
06
16
08
18
01
11
03
13
05
15
07
17
09
P2
P1
35 m
(b) 50 runs: set-up x state (cell text = day, Sep 2020)
SM-Y-P1
SM-Z-P1
SM-Y-P2
SM-Z-P2
NM-Z-P2
U1
U2
1
2
3
4
5
6
7
8
state (U = undamaged repeat, 1-8 = DS1-DS8)
air temperature (°C)
(c) Frequency-domain decomposition: 17 reference modes (numbered) from the 10 undamaged runs
first singular value / median
frequency (Hz)
lateral sweep, P1 (UDS)
vertical sweep, P2 (UDS)
vertical random, P2 (UDS)